

Title Page for Journal of Financial Market Infrastructures Submission

Manuscript Title

Demurrage-Based Monetary Circulation in Permissioned Financial Market Infrastructure: A Gesell-Inspired Approach to Settlement Efficiency

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Running Header/Short Title

Demurrage in Financial Infrastructure (39 characters)

Manuscript Statistics

- Word count: 8,742 words
- Number of figures: 0
- Number of tables: 0

Keywords

Demurrage, financial market infrastructure, settlement efficiency, reputation-based governance, distributed ledger technology, OTC derivatives

Code Availability

The implementation code and simulation framework referenced in this paper are available in the public GitHub repository: <https://github.com/Dylanthoma5667/spiralblockchain>

JEL Classification

G20 - Financial Institutions and Services

Abstract

This paper presents a novel integration of Silvio Gesell's demurrage-based currency theory with contemporary distributed ledger technology for regulated financial market infrastructure. We propose a dynamic demurrage mechanism for permissioned settlement systems that prevents the pathological accumulation of token value while maintaining network efficiency. Through formal analysis and simulation, we demonstrate that time-decay of idle holdings naturally incentivizes productive participation, dynamic demurrage rates scale with network inactivity creating adaptive disinflation, and Byzantine-resistant verification protects the system against malicious actors. We validate our approach through reputation-weighted validation simulations, showing 90% resilience against false utility claims while maintaining efficient block synchronization across institutional networks. The integration of Gesell's economic principles with modern cryptographic consensus achieves a purchasing-power-stable system that scales infinitely without hyperinflation, offering significant potential for OTC derivatives settlement and reconciliation workflows.

Key Messages

- Demurrage enhances circulation in permissioned settlement systems
- Reputation-based governance improves trust and operational resilience

- Dynamic monetary policies adapt to network conditions without inflation
- Byzantine fault tolerance ensures security in regulated environments